Answer Key & Teacher Notes

Session 1 — Dividing a Decimal by a Whole Number · Session 2 — Dividing by a Decimal

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Dividing a Decimal by a Whole Number

TALK PROMPTS

- Why does the quotient's decimal point sit directly above the dividend's decimal point?
- What does checking by multiplying prove about your quotient?

PARTNER TASK

Quotient Check. One partner finds 50.4 ÷ 12 (Problem 1f); the other checks by multiplying the quotient by 12. Switch roles with 18.9 ÷ 7 (Problem 1e).

WATCH FOR

- Placing the decimal point by counting digits instead of lining it up above the dividend's point (36.8 ÷ 4 written as 0.92).
- Stopping at a remainder instead of annexing a zero and continuing (23.4 ÷ 8).
- Ignoring the decimal point entirely (8.4 ÷ 4 = 21).

SESSION 2 · TEACHER NOTES

Dividing by a Decimal

TALK PROMPTS

- What are you really multiplying by when you move both decimal points one place right?
- Why does the answer to 126 ÷ 9 also answer 12.6 ÷ 0.9?

PARTNER TASK

Rewrite and Check. One partner rewrites 14 ÷ 0.35 (Problem 1a) with a whole-number divisor; the other solves and checks by multiplying. Switch with 37.2 ÷ 3.1 (Problem 1b).

WATCH FOR

- Moving the decimal point in only one number (the divisor or the dividend), which changes the quotient by a power of 10.
- Using × 10 when the divisor has two decimal places (3.6 ÷ 0.12 rewritten as 36 ÷ 12).
- Expecting the quotient to be smaller than the dividend when the divisor is less than 1.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.B.3	1	2	1	2	2	1	2	2	2	1	1
S1 · Form B	6.NS.B.3	1	2	1	2	2	1	2	2	2	1	1
S2 · Form A	6.NS.B.3	1	2	1	2	2	1	2	2	2	1	1
S2 · Form B	6.NS.B.3	1	2	1	2	2	1	2	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

dividend: The number being **divided**; the amount you share.

divisor: The number you **divide** by.

quotient: The **answer** to a division problem.

annex a zero: Write a 0 at the end of a decimal; the **value** stays the same.

- Put the decimal point in the quotient straight **above** the point in the dividend.
- If an amount is left over, **annex** a zero and keep dividing.
- Check: quotient × divisor = **dividend**.

Example: Write 25.2 ÷ 6 and put the point in the quotient straight **above** the point in 25.2. 6 goes into 25 **4** times with 1 left over. Bring down the 2 to make 12. 12 ÷ 6 = **2**, so the quotient is 4.2. Answer: Each piece is **4.2** m. Check: 4.2 × 6 = 25.2

YOU TRY

1. a. 2.1 b. 2.7 c. 3.75

2. $9.60 ÷ 4 = $2.40

EXIT TICKET

1. a. 9.3 b. 1.575 c. 3.24

2. 27.6 ÷ 8 = 3.45 kg (annex a zero: 27.60)

3. **B** (The answer is 5.25.)

PRACTICE FORM A

1. a. 3.9 b. 14.1 c. 3.75 d. 3.57 e. 2.7 f. 4.2

2. a. $96.75 ÷ 5 = $19.35 b. 43.2 ÷ 8 = 5.4 m (check: 19.35 × 5 = 96.75; 5.4 × 8 = 43.2)

3. 74.4 ÷ 6 = 12.4 m

4. Annex zeros (23.400) and keep dividing: 23.4 ÷ 8 = 2.925 m

5. $91.20 ÷ 16 = $5.70; Trey is not right (5.07 × 16 = 81.12)

6. $46.35 ÷ 5 = $9.27

7. a. 58.8 ÷ 7 = 8.4 m; 58.8 ÷ 14 = 4.2 m b. Twice as many pieces, so each piece is half as long.

8. a. $591.60 ÷ 6 = $98.60 b. 8 × $98.60 = $788.80 (check: 6 × $98.60 = $591.60)

9. He put the decimal point in the wrong place. The point goes straight above the point in 36.8, and 36 ÷ 4 = 9, so the answer is about 9: 36.8 ÷ 4 = 9.2.

10. **B** (3.2)

11. **A** (does not change its value)

Self-check word **DIVIDE**

PRACTICE FORM B

1. a. 3.15 b. 1.17 c. 3.75 d. 1.9 e. 5.28 f. 0.95

2. a. $83.70 ÷ 6 = $13.95 b. 37.8 ÷ 4 = 9.45 m (check: 13.95 × 6 = 83.70; 9.45 × 4 = 37.8)

3. 86.4 ÷ 6 = 14.4 m

4. Annex zeros (30.600) and keep dividing: 30.6 ÷ 8 = 3.825 m

5. $81.60 ÷ 16 = $5.10; Ava is not right (5.01 × 16 = 80.16)

6. $58.65 ÷ 5 = $11.73

7. a. 67.2 ÷ 7 = 9.6 m; 67.2 ÷ 14 = 4.8 m b. Twice as many pieces, so each piece is half as long.

8. a. $731.40 ÷ 6 = $121.90 b. 9 × $121.90 = $1,097.10 (check: 6 × $121.90 = $731.40)

9. She put the decimal point in the wrong place. The point goes straight above the point in 45.6, and 45.6 ÷ 6 is about 7, so 45.6 ÷ 6 = 7.6.

10. **C** (1.47)

11. **C** (straight above the point in 15.6)

Self-check word **TENTHS**

GUIDED NOTES

divisor: The number you **divide** by.

equivalent division: A new division problem with the **same** quotient.

power of ten: 10, 100, 1,000 and so on; each one is 10 times the **last**.

- Multiply the divisor and the dividend by the **same** power of 10 until the divisor is a **whole** number.
- The quotient does not **change**: 5 ÷ 0.25 = 500 ÷ 25 = 20.

Example: Write 8.4 ÷ 0.7. The divisor has **1** decimal place. Multiply both by 10: 0.7 becomes **7** and 8.4 becomes **84**. 84 ÷ 7 = 12. Check: 12 × 0.7 = 8.4 Answer: **12** bottles

YOU TRY

1. a. 64 ÷ 8 = 8 b. 27 ÷ 3 = 9 c. 180 ÷ 6 = 30

2. 96 ÷ 4 = 24 cups

EXIT TICKET

1. a. 315 ÷ 35 = 9 b. 78 ÷ 6 = 13 c. 330 ÷ 11 = 30

2. 156 ÷ 12 = 13 labels

3. **C** (greater than 6)

PRACTICE FORM A

1. a. 1,400 ÷ 35 = 40 b. 372 ÷ 31 = 12 c. 126 ÷ 9 = 14 d. 500 ÷ 25 = 20 e. 84 ÷ 2 = 42 f. 650 ÷ 5 = 130

2. a. 72 ÷ 6 = 12 beakers b. 45 ÷ 9 = 5 tubes

3. 117 ÷ 13 = 9 cans

4. 6.3 ÷ 7 = 0.9 is 10 times too small. Correct: 63 ÷ 7 = 9

5. 204 ÷ 12 = 17 bags

6. 91 ÷ 7 = 13 tails

7. She must multiply both by 100: 3.6 becomes 360, not 36. Correct: 360 ÷ 12 = 30

8. a. 126 ÷ 6 = 21 doors b. 126 ÷ 9 = 14 doors c. The same paint is shared in bigger portions, so fewer portions fit.

9. 56 ÷ 8 = 7, a whole number; Oren put a decimal point back in. The equivalent problem has the same quotient, so 5.6 ÷ 0.8 = 7.

10. **B** (9)

11. **B** (100)

Self-check word **DECIMAL**

PRACTICE FORM B

1. a. 240 ÷ 15 = 16 b. 975 ÷ 25 = 39 c. 192 ÷ 8 = 24 d. 225 ÷ 15 = 15 e. 54 ÷ 9 = 6 f. 88 ÷ 4 = 22

2. a. 84 ÷ 7 = 12 pots b. 56 ÷ 8 = 7 trays

3. 143 ÷ 11 = 13 cans

4. 7.2 ÷ 9 = 0.8 is 10 times too small. Correct: 72 ÷ 9 = 8

5. 306 ÷ 18 = 17 bags

6. 112 ÷ 8 = 14 tails

7. She must multiply both by 100: 4.2 becomes 420, not 42. Correct: 420 ÷ 12 = 35

8. a. 168 ÷ 8 = 21 doors b. 168 ÷ 12 = 14 doors c. The same paint is shared in bigger portions, so fewer portions fit.

9. 48 ÷ 6 = 8, a whole number; Malik put a decimal point back in. The equivalent problem has the same quotient, so 4.8 ÷ 0.6 = 8.

10. **D** (9)

11. **C** (equivalent)

Self-check word **DIVISOR**

SHEET 1 · DIVIDING A DECIMAL BY A WHOLE NUMBER

1. a. 3.2 b. 3.7 c. 0.42 d. 0.07 e. 5.2 f. 1.55
2. $16.80 ÷ 12 = $1.40 per pound (check: 1.40 × 12 = 16.80)
3. 10.2 ÷ 8 = 1.275 km (annex two zeros: 10.200)
4. a. $3.56 ÷ 4 = $0.89 b. $5.10 ÷ 6 = $0.85 c. Store B (4 cents less per can)
5. 4.35 ÷ 5 must be less than 1 because 4.35 < 5. Marcus put the decimal point in the wrong place (the point goes straight above the point in 4.35). Correct quotient: 0.87.

SHEET 2 · DIVIDING BY A DECIMAL

1. a. 48 ÷ 6 = 8 b. 27.5 ÷ 5 = 5.5 c. 96 ÷ 12 = 8 d. 90 ÷ 3 = 30 e. 168 ÷ 4 = 42 f. 132 ÷ 11 = 12
2. 7.5 ÷ 1.25 = 750 ÷ 125 = 6 pillows
3. a. 2.4 ÷ 0.3 = 24 ÷ 3 = 8 glasses b. 2.4 ÷ 0.2 = 24 ÷ 2 = 12 glasses (smaller glasses, so more of them)
4. a. 4.5 ÷ 0.75 = 450 ÷ 75 = 6 laps b. 6 ÷ 0.75 = 600 ÷ 75 = 8 laps
5. She moved the divisor's point two places (0.08 to 8) but the dividend's point only one place (2.4 to 24). Both must move two places: 240 ÷ 8 = 30.
6. **B** (67.2 ÷ 8)
7. **C** (39)

October 2026 conceptual additions — teacher key

Session 1 · Dividing a Decimal by a Whole Number

72 tenths ÷ 3 = 24 tenths = 2.4 meters. An estimate is 6 ÷ 3 = 2, so 2.4 is reasonable. Check 2.4 × 3 = 7.2.

Extension: Each share must be less than the original 7.2 meters when dividing it into three positive equal shares; 24 exceeds the total.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Dividing by a Decimal

4.8 L is 48 tenths and 0.6 L is 6 tenths, so 48 ÷ 6 = 8 bottles. Scaling both numbers by 10 preserves their ratio; scaling only one changes it. Check 8 × 0.6 = 4.8.

Extension: The quotient counts bottles smaller than one liter. Dividing by a positive amount less than 1 can produce a larger numerical quotient.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.